

Knapsack Problem

Knapsack Problem

You are given the following-

- A knapsack (kind of shoulder bag) with limited weight capacity.
- Few items each having some weight and value.

The problem states-

Which items should be placed into the knapsack such that-

- The value or profit obtained by putting the items into the knapsack is maximum.
- And the weight limit of the knapsack does not exceed.

Fractional Knapsack Problem-

In Fractional Knapsack Problem,

- As the name suggests, items are divisible here.
- We can even put the fraction of any item into the knapsack if taking the complete item is not possible.
- It is solved using Greedy Method.

Fractional Knapsack Problem Using Greedy Method-

Fractional knapsack problem is solved using greedy method in the following steps-

Step-01:

For each item, compute its value / weight ratio.

Fractional Knapsack Problem (Cont..)

Step-02:

Arrange all the items in decreasing order of their value / weight ratio.

Step-03:

Start putting the items into the knapsack beginning from the item with the highest ratio. Put as many items as you can into the knapsack.

PRACTICE PROBLEM

- For the given set of items and knapsack capacity = 60 kg, find the optimal solution for the fractional knapsack problem making use of greedy approach.

Item	Weight	Value
1	5	30
2	10	40
3	15	45
4	22	77
5	25	90

Practice Problem (Cont..)

Step-01:

Compute the value / weight ratio for each item-

Items	Weight	Value	Ratio
1	5	30	6
2	10	40	4
3	15	45	3
4	22	77	3.5
5	25	90	3.6

Practice Problem (Cont..)

Step-02:

Sort all the items in decreasing order of their value / weight ratio-

I1	I2	I5	I4	I3
(6)	(4)	(3.6)	(3.5)	(3)

Practice Problem (Cont..)

Step-03:

Start filling the knapsack by putting the items into it one by one.

Knapsack Weight	Items in Knapsack	Cost
60	∅	0
55	I1	30
45	I1, I2	70
20	I1, I2, I5	160

Practice Problem (Cont..)

Now,

Knapsack weight left to be filled is 20 kg but item-4 has a weight of 22 kg.

Since in fractional knapsack problem, even the fraction of any item can be taken.

So, knapsack will contain the following items-

$\langle I_1, I_2, I_5, (20/22) I_4 \rangle$

Practice Problem (Cont..)

Total cost of the knapsack

$$= 160 + (20/27) \times 77$$

$$= 160 + 70$$

$$= 230 \text{ units}$$